Evaluating the usability of Google Classroom as an online learning portal for universities (Project Proposal)

AI Acknowledgment

This proposal was developed by the student. AI tools such as ChatGPT and Gemini LLM were used to assist with research organization and idea generation. Grammarly was used for proofreading and grammar correction. All data, interpretations, writing, and conclusions are original.

1.0 Introduction

1.1 Background:

Digital learning tools have become really important parts of today's education systems and they are considered a complete first solution for educational institutions to deliver materials and arrange an easy and effective communication between students and teachers. The digital learning platforms allowed teachers to share class materials and handle homework more easily and opened doors for all students around the world to study wherever they are without the need to attend physically. Among all the different digital learning platforms available, Google Classroom is considered one of the most popular platforms that schools and universities use. Many universities adopt and rely on Google Classroom as their main digital learning platform for two main reasons: how smoothly it works with other Google Workspace programs, and how simple it is for anyone to use it on whatever device they have near them. Whether the student has a big desktop computer, a laptop to carry around, a tablet, or just a phone. But despite how many schools and colleges around the world rely on it, the platform still has some problems with how easy it is to use which bothers students and teachers regularly. A lot of users say they have trouble in navigating in the platform. According to Google Classroom (2023) talking about challenges in Google Classroom, “some users have found it less intuitive or challenging to navigate, especially for those who are not familiar with Google's ecosystem.” Hence, the platform needs a detailed study about usability to figure out if Google Classroom actually helps achieve education goals like it should. This kind of review does several things at once: looking at how things work right now and also finding places where improvements could make users happier with their

experience. The main goal of this project is to check how usable Google Classroom is by using Human-Computer Interaction principles and proven methods that researchers have used successfully before in HCI studies. By applying these principles to the analysis, problems that keep happening when people use the platform frequently can be found. In this research, suggestions to make Google Classroom easier to understand and use so that it functions better and suits the teachers and students needs in their daily work will be provided, and the findings of this research will also help in identifying weaknesses and suggest solutions to minimize the challenges that students and teachers face regularly when using the platform and ensure an effective learning journey.

2.0 Aim:

The aim of this project is to evaluate how usable Google Classroom is as an online learning platform for the universities and identify weaknesses to improve using heuristic evaluation and the System Usability Scale method.

2.1 Objectives:

The project objectives are as follows:

1. Review existing literature and research papers on usability evaluation and the principles of human-computer interaction to strengthen the analysis.
2. Apply Nielsen's 10 usability criteria (Nielsen, 1994) to find out issues and evaluate Google Classroom's interface design.
3. Apply the System Usability Scale method.
4. Record the results including usability issues and document them through screenshots and detailed notes.
5. Suggest improvements to apply to enhance the learning journey

3.0 Methods of Solution

The project is going to use a heuristic evaluation approach, which is considered suitable to adopt for a single evaluator and it does not require a lot of evaluators to be involved. The Nielsen's 10 Usability Heuristics will be used because they help identify usability problems. Each key area of Google Classroom such as the Stream, Classwork, People, and Grades sections will be reviewed as part of the process and the Issues that are identified will be recorded in a table. The table will include the specific heuristic that was violated, a clear description of the problem and a severity rating that ranges from 0 (no issues) to 4 (needs to be resolved), and suggestions for improvements will be provided for each issue. The project will also follow and use the System Usability Scale to assign an overall score of the usability. The project will also include Screenshots that are captured throughout the evaluation process to document findings and provide evidence of the issues identified.

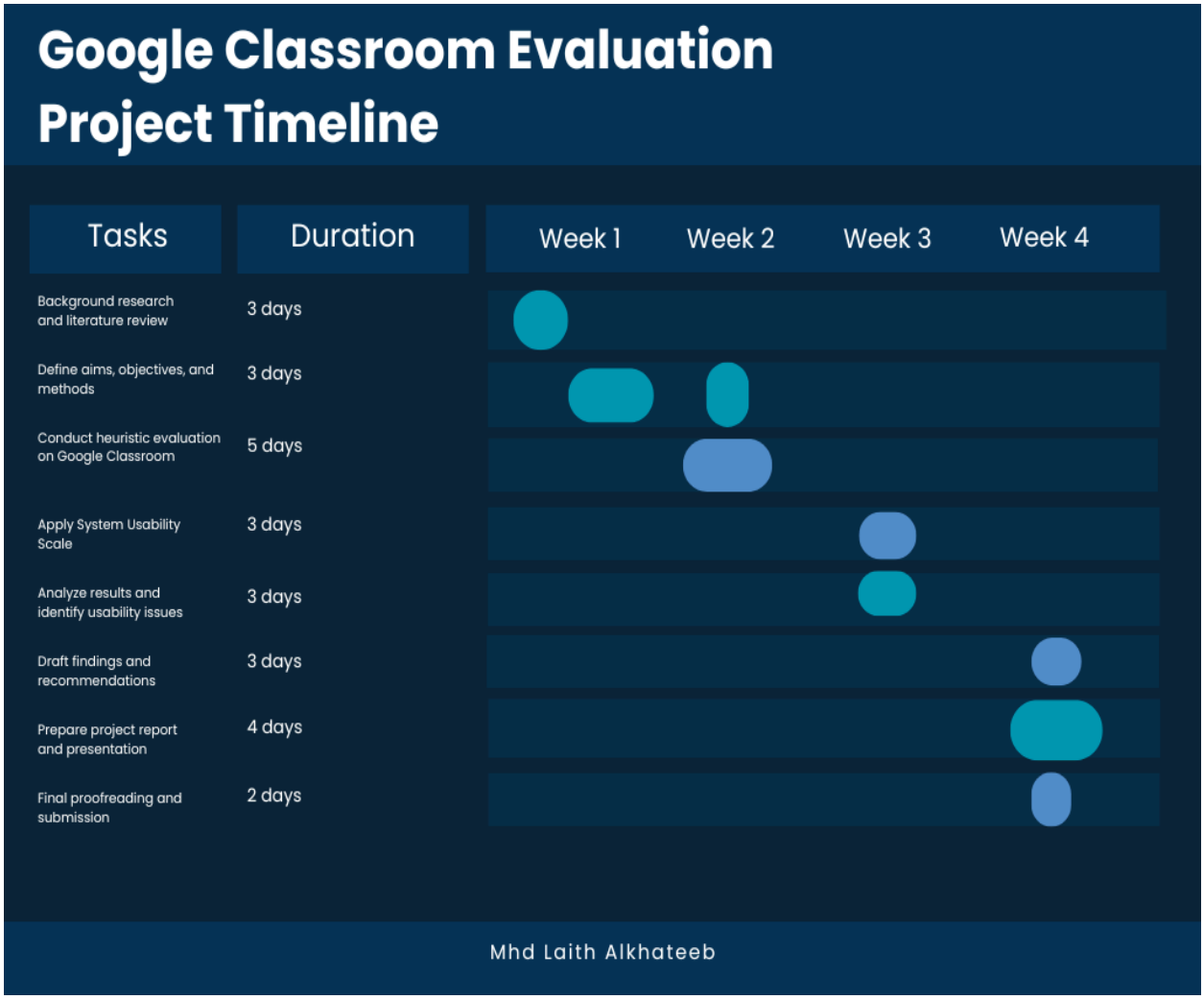
The analysis will focus on several key aspects including:

1. How accurate the information is presented
2. How consistent are the sections in the interface
3. How well the system prevents errors from occurring
4. How easy the user can navigate when moving through the platform.

4.0 Risk Management:

Risk	Impact	Likelihood	Action
Difficulties in time management	High	Medium	Develop a clear plan and assign objectives
Limited access to the features of the system	Medium	Low	Use an active student account to access all features
Self-evaluation bias	Medium	Medium	Follow the obvious and strict rules and document observations objectively.
Technical or internet problems	Medium	Low	Take offline notes and screenshots during the evaluation.

5.0 Project Timeline:



6.0 Project deliverables:

The expected deliverables will include:

1. A heuristic evaluation report will be presented as a table that contains the violated rules, severity ratings, and recommendations for each issue identified.

2. A completed System Usability Scale questionnaire will be provided along with a thorough score analysis.

3. Screenshots that highlight usability problems will be included to provide visual documentation of the issues.

4. Recommendations for improvements will be prepared and provided.

5. The final written report along with a PowerPoint Presentation that summarizes all the results of the research.

7.0 References:

Google Classroom (2023) 'What are the Major Challenges for Google Classroom', LinkedIn Pulse. Available at:https://www.linkedin.com/pulse/what-major-challenges-google-classroom-googleclassroom (Accessed: 11 November 2025).

Nielsen, J. (1994) '10 Usability Heuristics for User Interface Design', Nielsen Norman Group. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/ (Accessed: 11 November 2025).

Brooke, J., 1996. SUS-A quick and dirty usability scale. Usability evaluation in industry, 189(194), pp.4-7.

Evaluating the usability of Google Classroom as an online learning portal for universities (Project Report)

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1. Introduction

1.1 Background:

In today's education, online learning platforms have become vital tools for supporting the complete teaching and learning environment. These platforms improve communication between instructors and students while also allowing for resource sharing and streamlining assignment management processes. Among the different alternatives available, Google Classroom has emerged as one of the most widely utilized online learning portals in educational institutions worldwide. The platform's success may be attributed in large part to its flawless connection with Google Workspace, as well as the surprising simplicity with which users can access it across numerous devices. Whether students and teachers are using desktop computers, laptops, tablets, or cellphones, the accessibility is seamless. Despite its enormous popularity, the platform is not immune to usability issues that its users face regularly. Users commonly find challenges with navigation complications inside the interface (Google Classroom, 2023). To fully understand how effectively Google Classroom supports educational objectives, a comprehensive usability study is required. Such an evaluation not only analyzes present performance but also identifies key areas for improvement that would significantly improve the overall user experience. The major purpose of this project is to evaluate Google Classroom's usability from a Human-Computer Interaction perspective using well-established assessment methodologies that have been proven useful in HCI research. These analytical techniques will allow for the identification of specific weak spots that users repeatedly experience throughout their interactions with the platform. The findings of this research

will give practical recommendations for making Google Classroom more intuitive, efficient, and aligned with educational requirements.

2. Literature review

2.1 The Role of Online Learning Platforms in Higher Education

The rise of Distance learning and e-learning has completely changed the way information can be given to students by their instructors in modern education. Distance learning has become an essential part of learning systems in universities around the world. Students are no longer required to attend their regular classes physically because their regular courses can be given anywhere. All they need is a good device along with a good internet connection. The necessity of encouraging digital learning and presenting the idea of the "digital classroom" at the same time has been made easier by the development of educational technology.

Because nowadays, students are curious about how to learn using technology, educational institutions are being pushed to include digital learning into their teaching strategies. Because of this, educational organizations need a reliable system that is innovative, economical, and flexible.

The Learning Management Systems (LMS) concept was thus introduced as a solution. Learning management system is considered one of the most popular learning technologies in higher education, learning management systems (LMS), also known as virtual learning environments (VLE) or course management systems (CMS), are primarily responsible for facilitating digital teaching and learning. In addition to providing an infrastructure that facilitates communication, evaluation, collaboration, and the administration and management of learning materials, the learning management system (LMS) enables the use of a variety of educational resources. Among the well-known learning management systems utilized by educational institutions are Sakai, Fronter,

Google Classroom, Blendspace, Blackboard, Edmodo, Desire2Learn, and Moodle. (Kumar, Bervell, and Osman, 2020)

This range of systems mentioned that are all a part of the process are again covered under the name Learning Management System (LMS). These consist of e-learning platforms, learning portals, virtual learning systems, learning content management systems, learning portals, and virtual learning environments (LCMS, VLS, and VLE). There are three forms of presence used by the new LMS:

1. Instructional
2. social
3. cognitive.

While cognitive presence refers to the actual process of students gaining information through communication, social presence describes how students respond to the virtual learning community on a social and emotional level. (Samani, 2016)

The concept of e-learning and learning management systems has risen and been taken into account during the COVID-19 pandemic. Since then, the education sector has been completely revolutionized, leading to a mandatory integration of the emerging technologies related to education. According to (Fawareh and Al-Bardeen, 2021)

> Because of the pandemic of COVID-19 worldwide, people and governments have made great efforts to reduce the spread of COVID-19 through non-pharmaceutical interventions and protection measures that include social distancing and self-isolation in homes (UNESCO).
>
> Many critical sectors have been closed, including the education sector, as schools and university education in more than 100 countries around the world have been completely closed. Due to the closure of universities and schools, UNESCO has recommended the use of E-learning programs, including the opening of smart applications, educational platforms, and websites that students and lecturers can use for the continuity of the education process.
>
> E-learning appears as the best solution in the COVID-19 crisis as a pedagogy empowered by digital technology. In several modern fields, E-learning is viewed as a new form of learning that utilizes the advantages of the Internet to create quality and interactive instructional materials and services. E-learning has been going on for decades in many countries around the world (Nicholson, 2007). E-learning offers the ability to share material in all kinds of formats, such as videos, slideshows, Word documents, and PDFs. The main aim of E-learning systems is to improve the whole educational system and to enhance the

> interaction between students and teachers (Cohen, 2010).” (Fawareh and Al-Bardeen, 2021, p. 1)

Furthermore, Mohammad Alshira’h, Mohammad Al-Omari, and Bashar Igried (2021) point out that “The outbreak of the COVID-19 pandemic has forced countries around the world to close down educational institutions at all levels. (Mohammad Alshira’h et al., 2021, p. 6431)”

These statements lead that the pandemic back then has strengthened the need to adopt the concept of e-learning in the education sector.

To deep dive into the concept of e-learning, Mohammad Alshira’h, Mohammad Al-Omari, and Bashar Igried (2021) defined e-learning as

> An advanced educational system that combines the concept of distance learning and traditional learning. It can support and manage the learning process remotely without students being present on campus. E-learning systems provide both students and instructors with tremendous tools that facilitate the learning and teaching process [1]. These systems provide students with educational services, such as granting them access to the academic materials that are provided as online materials (i.e., slides, supporting documents). Students can download previously recorded lectures. Moreover, the e-learning system allows students to communicate with each other and with the instructor through the academic forums and chat tools ([15], [16]).

By using various e-learning platforms like Moodle, Google Classroom, and Blackboard, the concept of e-learning has been fully acknowledged. Additionally, fully interactive lectures and meetings are conducted using additional platforms like Zoom and Microsoft Teams. At the end of each semester, final tests can be conducted at locations chosen by the university or online using systems designed to enable remote monitoring by digital cameras, according to certain requirements and restrictions.

Students can access course materials, presentations, files, and other electronic information all in one place. These are among the most crucial educational features offered by e-learning platforms. In addition to offering ways for students to connect with lecturers through interactive discussion boards and direct messaging, e-learning platforms allow students to download and view lectures that have been recorded. Additionally, students can also gain access to a variety of educational services, technical support, and help through e-learning platforms. Consequently, instruction occurs off-campus. (Mohammad Alshira'h, Mohammad Al-Omari, and Bashar Igried, 2021)

2.2 Review of Google Classroom as an Online Learning Platform (OLP)

Among the top Learning management systems (LMSs) that benefit both teachers and students and increase the interaction between both aspects comes Google Classroom.

Google created the Google Classroom (GC) platform to assist educators in creating and overseeing online courses. It is considered a useful tool for both teachers and students because it is user-friendly, free to access, and it is an easy setup and accessibility, whether on a computer or smartphone with an internet connection. In addition to helping teachers carry out their lesson strategies, Google Classroom (GC) also facilitates effective student engagement, which raises comprehension and involvement in class activities.

In other words, Google Classroom (GC) can be defined as a cloud-based learning management system (LMS) that uses technology to track student engagement, produce material, provide instruction, and evaluate student achievement.

Google Classroom (GC) allows students to advance at their own pace and provides teachers with the ability to track their progress, highlight areas in which students are having difficulty, and modify their teaching strategies appropriately, in addition to its features. It also supports personalized learning. (Al Shabibi, 2025)

Studies (cited in Al Shabibi, 2025) mentioning how effective Google Classroom (GC) is from the teacher's side have shown that:

1. Google Classroom has been found and proven as a valuable tool for teachers when it comes to delivering and distributing e-learning, especially during the pandemic and the shuffle of the teaching methods that happened back then.
2. Google Classroom-supported learning was found to be valid, practical, and effective in Blended learning environments.
3. Teachers use Google Classroom (GC) to design structured lessons, implement support activities, and provide problem-solving processes.
4. Google Classroom (GC) simplifies tasks such as assignment management and communication.

On the other hand, from the students' perspective, studies have shown that:

1. Google Classroom (GC) has significantly improved mathematics learning outcomes for high school students by enabling learning both inside and outside the classroom. It also improved reading and writing skills for students in the Arab world.

2. Students found Google Classroom (GC) to be a successful and effective online learning environment that helped them develop knowledge, evaluate performance, and provide feedback.

3. Students found Google Classroom (GC) to be engaging and non-boring as a medium. It encourages teamwork, allows students to share ideas, and facilitates comfortable communication.
4. Students appear to express high satisfaction with Google Classroom (GC). Viewing it as a valuable support tool. Features like its ease of use, mobile-friendly design, and time-saving features made it popular. Studies have also shown that over 90% of students rated it "very good" or "good”.

However, Al Shabibi (2025) pointed out that some challenges were noted regarding accessibility. Tools such as wikis in virtual classrooms, in particular. (Al Shabibi, 2025)

While the positive offering of Google Classroom had been presented. Al Shabibi (2025) argued that Challenges still exist.

Key advantages of Google Classroom (GC) are as follows:

1. Google Classroom (GC) is considered a highly useful facilitation tool for teachers to manage assignments, announcements, and assessments online.

2. Google Classroom (GC) shows better organization when it comes to the classroom, and it is highly valued by students for its accessibility and organizational features.

3. Improved teacher-student interaction is also part of Google Classroom features. As it enables teachers to communicate with students even outside of the physical classroom, which enhances overall engagement.

4. Students' intentions of continuing to use Google Classroom come from the high satisfaction and the ease of use of the platform. Which is critical for all Learning Management Systems (LMSs).

Key challenges of Google Classroom are as follows:

1. From teachers' perspectives, teachers noted that the Google Classroom interface is somewhat complicated and lacks user-friendliness. This could affect the overall efficiency of the platform.

2. Teachers believe that the platform is more of a document and a classroom management tool rather than a transformative teaching methodology. These findings are limiting the capabilities and potential benefits of the platform. As teachers had not fully explored the platform's features, as per the studies found (cited in Al Shabibi, 2025)

3. Teachers expressed concern about students misusing mobile devices during class, which can lead to potential distractions. (Al Shabibi, 2025)

These findings lead to reconsidering the platform's usability according to HCI Principles and Usability principles is a must.

2.3 Fundamentals of Human-Computer Interaction (HCI)

In order to improve and increase system capabilities and meet the needs and necessities of users, usability and HCI are increasingly becoming essential components of the system development process. While usability will verify whether the system is efficient, effective, safe, useful, easy to learn, easy to remember, easy to use, and evaluate, practically visible, and give users job satisfaction, HCI will help designers, analysts, and users identify the system's needs from text style, fonts, layout, graphics, and color. By integrating these elements into the system development process, such as the sustainable design, users' objectives and tasks will be measured and completed through the use of a particular technology. (Issa and Isaias, 2022)

According to Issa and Isaias (2022),

> In order for computer-based systems to be widely accepted and used effectively, they need to be well designed via a “user-centered” approach. This is not to say that all systems have to be designed to accommodate everyone, but that computer-based systems should be designed for the needs and capabilities of the people for whom they are intended.

This statement concludes that it is not necessary to design every system to meet everyone's needs. But to design the systems by adhering to a particular criterion, targeted people, and the field in which a specific system is dedicated to. This means that in the education sector. Every Learning Management system should be dedicated

to the people who are using it, such as educators and students. Making it more user-friendly and easy to use for both perspectives. (Mhd Laith Alkhateeb, 2025)

The phrases "interface" and "interaction," which are associated with the development process, should be defined before exploring deeply into the subject of human–computer interaction. As stated by a study (cited in Issa and Isaias, 2022), the interface is the "visible piece of a system that a user sees, hears, or touches." A broader phrase that includes user activity is interaction. For example, an interaction occurs when a user clicks with a mouse or types something on the keyboard. The design of websites is subject to the broad principles of HCI. (Issa and Isaias, 2022).

In the middle of the 1980s, the phrase "human–computer interaction" (HCI) was introduced. (Issa and Isaias, 2022).

Designing, assessing, and implementing interactive computer systems for human use, as well as researching the key phenomena that surround them, are the focus of the field of human-computer interaction (HCI). (Issa and Isaias, 2022).

HCI is concerned with designing computer systems that are not only functional but also safe, effective, simple, and enjoyable to use. (Issa and Isaias, 2022).

Furthermore, HCI is focused on the effects of new methods like speech recognition and generation (input and output), as well as how current input and output technologies affect interaction (Issa and Isaias, 2022).

The field of human-computer interaction (HCI) is concerned with the development, assessment, and deployment of interactive computer systems. The system should be simple for users to use in their workplaces if the system were developed using HCI ideas and techniques. (Issa and Isaias, 2022).

In other words, HCI can be defined as:

> A field of study that focuses on improving computer interfaces to make computer systems simple and easy to use. It can be said to be the combining of human user and computer system with the intention of performing a task involving hardware and software components [5]. The HCI occupies a specific point in the intersection between the social and behavioral sciences, as well as between computer and information technology [6]. Improving the interaction between users and computers is the basic goal of HCI. This is achieved by developing appropriate systems that are user-friendly, efficient, and meet the needs of users [7]. Another fundamental goal of HCI is to improve the usability, utility, safety, effectiveness and efficiency of systems that include computers. (Al-Dossari, 2017)

Issa and Isaias (2022) clearly defined the goals of HCI and its purpose, emphasizing that “The goals of HCI are to produce usable and safe systems, as well as functional systems. These goals can be summarized as safety, utility, effectiveness, efficiency, and appeal. These goals focus on the services that the system provides, how quickly the tasks can be achieved, and ensuring that users like the system. In general, usability is an essential concept in HCI and is concerned with making systems easy to learn, easy to use, and with limiting error frequency and severity.”

“The purpose of HCI is to design a computer system to match the needs and requirements of the users.” (Issa and Isaias, 2022)

From an e-learning perspective, successful learning requires the use of e-learning technologies that use pedagogical techniques and HCI principles. E-learning becomes a crucial component of educational institutions that encourage students and facilitate their engagement. E-learning facilitates quick comprehension and allows for flexible interaction between the learner and the materials. Additionally, e-learning helps the disabled community in meeting learning goals in a flexible and consistent manner. Additionally, it allows students to learn at their own pace and accomplish whatever they need to. Designers of e-learning must make sure that it accommodates a wide range of individual preferences and talents rather than excluding or restricting any one set of users. (Al Mahdi, Rao Naidu, and Kurian, 2019)

Regardless of unclear circumstances, the interface must enable the learner to retrieve information. In order to reduce the risks associated with interface design and the effects of unintentional or accidental behaviors, the learner must be assisted. An e-learning application's interface should help the user learn effectively, pleasantly, and with the least amount of tiredness. The following requirements should be met by the design in order to meet the expectations of both the teacher and the students.

1. The subject matter must have a pleasing visual look.

2. Clear instructions on how to use the program

3. Learning must be made enjoyable through animations and images.

4. The navigation structure is easy to use. (Al Mahdi, Rao Naidu, and Kurian, 2019)

2.4 Usability and User Experience (UX) Concepts

One essential component of human-computer interaction is usability. Originally used to refer to efficiency, this phrase was later improved to include two metrics: effectiveness and usability. The last type assesses the interface, whereas the former analyses user performance. Nielsen then presented ten usability principles in 1994. (Al-Dossari, 2017)

According to the International Standard Organization ISO 9241-11 (cited in Al-Dossari, 2017), usability is the degree to which a product can be used by specific users to accomplish specific goals in a specific context of usage with effectiveness, efficiency, and satisfaction.

According to Baecker and Grudin (cited in Al-Dossari, 2017), usability is a wide notion that essentially relates to how simple a system is for users to learn, how effective it is once they have mastered it, and how enjoyable it is to use. (Al-Dossari, 2017)

In the following questions, Nielsen (cited in Al-Dossari, 2017) defines usability using five quality components and claims that the term relates to techniques for enhancing usability during the design process:

1. Learnability: To what extent can computer users do simple activities when they first come across the design?

2. Efficiency: After mastering the design, how fast can users complete tasks?

3. Memorability: After a break, how simple is it for users to regain proficiency?

4. Errors: How frequently do users make mistakes, how serious are they, and how quickly can they fix them?

5. Satisfaction: To what extent are people happy with the design? (Al-Dossari, 2017)

In the context of e-learning, Usability plays an important role in the overall presentation of the platform. Because Student interactions with e-learning interfaces should lead to actual learning rather than task fulfillment. (Ssemugabi, 2010)

Usability has a substantial impact on learning. An e-learning product is both instructional content and a computing system. Therefore, a resource's usability includes both its learning efficiency and users' satisfaction. (Ssemugabi, 2010)

According to Abuhlfaia and De Quincey (2019)

> Usability has an essential role in the acceptance and effectiveness of e-learning platforms. If usability is poor, students will spend time becoming comfortable with system functionality at the expense of learning [16]. However, the usability of e-learning platforms is a challenge for designers [17] and has been shown to have a substantial impact on students [18].” (Abuhlfaia and De Quincey, 2019, p.

Dewi (2023) also stated that “the effectiveness of online learning platforms heavily relies on their usability and user experience” (Dewi, 2023). This emphasizes the importance and the strong influence of Usability on online learning platforms. Including the effectiveness and the ease of use.

The following table demonstrates key findings of prior studies conducted on Google Classroom as an online learning platform (OLP). Each study used a different evaluation method to assess the effectiveness and the ease of use of Google Classroom, such as the System Usability Scale (SUS), Heuristic evaluation, and online surveys.

Author(s) & Year	Study Title	Evaluation Method(s)	Participants / Sample	Key Findings (Usability Issues / Strengths)	Reported SUS Score / Metrics
Al Shabibi (2025)	Teachers' Perception of Effectiveness of Google Classroom Using the System Usability Scale (SUS)	Descriptive approach (Quantitative SUS survey + Qualitative semi-structured interviews)	217 male and female teachers from the education directorates in Oman	Advantages: 'Good' was the rating for perceived effectiveness. Teachers discovered that it had strong platform tools (assessment, organization), stimulated creativity, met tech vision, and simplified the educational process (less paper/cost). Students couldn't wait to use it. Problems/Difficulties: 'Learning loss', lack of preparation by the Ministry of Education, preference for in-person instruction, and poor learning. Teachers of maths and science faced challenges (inverted letters, absence of symbols). Some icons are missing on mobile devices, and there are platform issues (black	69.2 (good)

				screen, voice vanishing). Problems with the infrastructure (bad internet connection, no devices).	
Reddivari and Gattupalli (2021)	Comparative study of Canvas and Google Classroom Learning Management Systems using usability heuristics	Survey evaluation based on usability heuristics	9 users (students), who are mostly familiar with Canvas	Strengths of Google Classroom compared to Canvas: Most people considered registration to be easier (55.6%), and the process of uploading a photo was considered to be more similar to other apps. 7 out of 10 usability heuristics were satisfied. Problems (GC compared to Canvas): Canvas provides better submission status notifications. The heuristic "Match between system and real world" was broken, making "Stream" unclear. Most users found it difficult or impossible to message other users (violating the recognition heuristic rather than the recall heuristic). The Consistency and Standards Heuristic was partially met.	Mean satisfaction ratings on a 1-10 scale were converted to metrics: Variance = 1.55, Standard Deviation = 1.247.
	Usability and satisfaction of Google Classroom as an instructional teaching and	Online survey	128 students (75% female, 25% male) from a State Islamic Institute in Kendari,	Strengths: Platform of choice because it is visually appealing and user-friendly (41.4%). 95 percent of students expressed that they	(Focus on satisfaction levels throughout the Likert scale. 57%

(Alim *et al.*, 2021)	learning medium: the students' perspective		Indonesia	were extremely pleased or fairly satisfied. Benefits include mobile-friendly design, paperless operation, and simplicity of management. Mostly used for schoolwork, particularly homework (89.8%). Problems: Internet cost (32.8%) and connection (64.8%) were the main obstacles.	were "Quite satisfied," 21.1% "Satisfied")
Dewi (2023)	USABILITY ANALYSIS OF HUMAN COMPUTER INTERACTION IN GOOGLE CLASSROOM AND MICROSOFT TEAMS	Quantitative descriptive analysis using a survey (HCI usability The indicators are: Learnability, Efficiency, Memorability, Errors, and Satisfaction.	494 respondents (students) from educational institutions in the Greater Jakarta Area	Advantages of GC compared to Microsoft Teams: improved results on every usability metric. Considered to be more efficient (55.83% high score), easier to learn (70.41% high score), more remembered (96.25% high score), and having a lower error rate (29.58% low error rate). Better overall satisfaction (high score of 58.75%). Simpler navigation, a more straightforward layout, and smooth Google app integration. Problems: Accessing and enrolling in classes continued to be difficult. Assignment submission notifications are delayed (both platforms share this issue).	Mean percentage scores reported for Google Classroom (GC): Learnability: 49.48%, Efficiency: 30.69%, Memorability : 35.71%, Errors: 18.94%, Satisfaction: 26.45%

Kumar, J.A., Bervell, B., and Osman, S. (2020)	Google Classroom: Insights from Malaysian higher education students' and instructors' experiences	Qualitative study (semi-structured interviews)	17 students and 3 teachers from a higher education institution in Malaysia (beginner GC users)	Strengths: The primary elements were usefulness and simplicity of use. Turning in the assignment was the simplest and most beneficial. Mobile accessibility, reminders, and notifications were considered excellent. Supports paperless classrooms, which save money and time. Teachers appreciated how simple it was to share and manage information, save files on the drive automatically, and integrate other programs. Problems/Difficulties: Students: It was annoying to scroll for previous postings using Facebook's "stream" interface. Simple colors and a dull UI (no personalization or aesthetics). Absence of two-way peer/teacher engagement and communication tools (private messaging, online presence monitoring). Classwork is not private because others may see and comment on it. Poor internet access and cost. Teachers: dull user interface. Google Drive storage is limited, and learning metrics	(Qualitative focus)

				are not available for free accounts. Validating students in class is challenging since anybody with code may join.	

3. Research Methodology

3.1 Research Design and Justification

For Google Classroom, this study will adopt mixed methods: qualitative and quantitative. Aiming to assess the usability of Google Classroom as an online learning platform (OLP) (Mhd Laith Alkhateeb, 2025). These methods are:

1. Jakob Nielsen's usability heuristics method

2. System Usability Scale (SUS)

These methods are considered reliable, easy to implement, and free. These methods will be applied from a student perspective as a solo evaluation process. Although the existence of participants in the evaluation process will strengthen the evaluation procedure and final results. The process was carried out individually, as no participants were available. According to Paramitha, Dantes, and Indrawan (2018), mentioning the processes of the evaluation using usability heuristics by Jakob Nielsen (cited in Paramitha, Dantes, and Indrawan, 2018), "Heuristic evaluation does not demand a representative number of people."

Nielsen (1995) added that 3-5 people are usually enough to conduct a Heuristic analysis. Moreover, he also added that even one person can figure out a usability issue. (Nielsen, 1995)

3.2 Heuristic Evaluation Method

Jakob Nielsen created the heuristic evaluation method to assess the Usability of a website or a platform. According to Paramitha, Dantes, and Indrawan (2018),

> The purpose of a heuristic evaluation is to improve the interface design effectively. The evaluators then evaluate the interface based on predetermined heuristic principles. If there is a violation of the heuristic principle, the evaluator decides the severity of the problem, then provides a recommendation for the problem. Problems which detected at the design stage can be fixed before the system implementation stage. Heuristic evaluation is an excellent technique for design interface evaluation, since it is easier to find or determine the problems that arise.

The Usability Heuristics criteria is made of 10 usability principles. The principles are as follows:

> “1. Visibility of system status: The system should always keep users informed about what is going on, through appropriate feedback within a reasonable time.
>
> 2. Match between system and the real world: The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.

3. User control and freedom: Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

4. Consistency and standards: Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

5. Error prevention: Even better than good error messages is a careful design that prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

6. Recognition rather than recall: Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

7. Flexibility and efficiency of use: Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8. Aesthetic and minimalist design: Dialogues should not contain information that is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9. Help users recognize, diagnose, and recover from errors: Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10. Help and documentation: Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large." (Jakob Nielsen, 2005)

For an evaluation of a Learning Management System, Mtebe and Kissaka (2015) have proposed a refined version of Jakob Nielsen's Usability Heuristics to meet the requirements of Learning Management Systems.

The refined version is as follows:

> "1. Visibility of System Status: The system should ensure that learners are informed about what is happening through appropriate feedback within a reasonable time (Albion, 1999; Jakob Nielsen, 1994). The system should also provide a visual or audio response so that learners can understand the result of their actions (Squires & Preece, 1999).
>
> 2. Match Between the System and the Real World: The system should ensure that the LMS uses learners' language, with words, phrases, and concepts familiar to the learner rather than system-oriented terms (Jakob Nielsen, 1994). Moreover, system information should be presented and organized in a natural and logical order (Jakob Nielsen, 1994; Squires & Preece, 1999).
>
> 3. User Control and Freedom: The system should have facilities that enable the learner to leave the system anytime through exit signs, undo and redo operations (Jakob Nielsen, 1994).
>
> 4. Consistency and Standards: The system should ensure that the learner experiences the user interface as consistent in terms of menus, color, typography, and dialog design (Alsumait & Al-Osaimi, 2010). However, the learning resources are varied.

5. Error Prevention: The system should be carefully designed to prevent common errors from occurring in the first place (Jakob Nielsen, 1994). However, it should have procedures to enable learners to fix errors when they appear. Moreover, the system should prevent learners from making irreversible errors (Jakob Nielsen, 1994).

6. Recognition Rather than Recall: The system should minimize the learner's memory load by making objects, actions, and options visible, and the learner should not have to remember information from one part of the dialogue to another (Jakob Nielsen, 1994).

7. Flexibility and Efficiency of Use: The system should cater to different levels of users, from novice to expert, with expert users given the opportunity to use shortcuts and other advanced tools (Albion, 1999; Jakob Nielsen, 1994).

8. Authenticity and Minimalism in Design: The system dialogues must not consist of irrelevant or rarely needed information that may distract learners during the learning process (Jakob Nielsen, 1994).

9. Recognition, Diagnosis, and Recovery from Errors: The system should express error messages in simple language that does not include programmer code, precisely indicate the problem, and in a friendly way suggest a solution that the learner can handle (Alsumait & Al-Osaimi, 2010; Jakob Nielsen, 1994).

10. Help and documentation: The system should provide appropriate online help and documentation that is easily accessed, easy to search, and related to the

learners' needs (Albion, 1999; Alsumait & Al-Osaimi, 2010; Jakob Nielsen, 1994)." (Mtebe and Kissaka, 2015, p. 3)"

(Mtebe and Kissaka, 2015)

3.3 System Usability Scale (SUS) Method

One of the most popular evaluation methods is the System Usability Scale. System Usability Scale (SUS) is a popular standardized tool for evaluating perceived usability. According to research (cited in Lewis, 2018), 43% of post-study questionnaire usage in industrial usability studies was attributed to the SUS. The work that introduced the SUS had 5,664 citations according to Google Scholar. The SUS consists of 10 five-point items with alternating positive and negative tones in its standard (most frequently used) version. (Lewis, 2018)

According to Alkinani and Alzahrani (2021), “The SUS is simple and inexpensive compared to many other methods of system usability evaluation. Moreover, the questionnaire can be completed quickly because it is short and contains only 10 statements. The SUS score is between 0 and 100, which makes it understandable by experts and non-experts.”

The use of human factors psychology in the planning and assessment of personal computer and office systems increased dramatically in the early 1980s. It became evident that evaluating perceived usability was just as essential as objective usability (effectiveness and efficiency) for commercial computer products. (Lewis, 2018)

There were a number of standardized questionnaires already in place for assessing user satisfaction with systems, but they were not made to evaluate usability after task-based usability studies. In the late 1980s, a number of researchers—some created in universities, others at large corporations—published standardized usability surveys, hoping to close that gap. (Lewis, 2018)

Several of them are still in use, such as The University of Maryland, College Park's Questionnaire for User Interaction Satisfaction (QUIS), University College, Cork's Software Usability Measurement Inventory (SUMI), International Business Machines Corporation's Post-Study System Usability Questionnaire (PSSUQ) and its non-lab version, the Computer Systems Usability Questionnaire (CSUQ), and Digital Equipment Corporation or (SUS). The SUS may have been the first of these to be created, but it was the last to be published. (Lewis, 2018)

According to Lewis (2018), the creator of the SUS (John Brooke) wrote about it by saying:

> “[In 1984] as part of a usability engineering program, I developed a questionnaire – the System Usability Scale (SUS) – that could be used to take a quick measurement of how people perceived the usability of computer systems on which they were working. This proved to be an extremely simple and reliable tool for use when doing usability evaluations, and I decided, with the blessing of engineering management at Digital Equipment Co. Ltd (DEC; where I developed SUS), that it was probably something that could be used by other organizations (the benefit for us being that if they did use it, we potentially had something we could use to compare their systems against ours). So, in 1986, I made SUS freely available to a number of colleagues, with permission to pass it on to anybody else who might find it useful, and over the next few years occasionally heard of evaluations of systems where researchers and usability engineers had used it with some success. Eventually, about a decade after I first created it, I contributed a chapter describing SUS to a book on usability engineering in

> industry (Brooke, 1996). Since then, it has been incorporated into commercial usability evaluation toolkits such as Morae, and I have recently seen several publications refer to it as an "industry standard" – although it has never been through any formal standardization process." (Lewis, 2018, p. 577)

Lewis (2018) also added about the SUS that "There are no fees required to use the SUS. "The only prerequisite for its use is that any published report should acknowledge the source of the measure" (Brooke, 1996, p. 194). Thus, researchers who use the SUS should acknowledge Brooke (1996) as the source in internal reports and external publications."

And, "There are no statistical tests for this type of validity because it depends on rational rather than empirical assessment of the source of the items. Creation of items by domain experts or selection from a literature review of existing questionnaires in the target or related domain are typically taken as evidence supporting the claim of content validity." (Lewis, 2018)

In terms of calculating the System Usability Scale (SUS), the various tones of the items and the early choice to adjust the score to range from 0 to 100 make the traditional method of scoring the SUS quite complex. The initial scoring step conceptually involves converting raw item scores into adjusted scores (also called "score contributions") that are different for odd- and even-numbered items (items with positive and negative tones), ranging from 0 (poorest rating) to 4 (best rating). (Lewis, 2018)

Since all 10 questions must be rated according to the SUS scoring system, an item that is left blank by a respondent should receive a raw score of 3, which is the middle of the five-point scale. Subtract 1 from the raw score for items with odd numbers, and 5 from the raw score for items with even numbers. To obtain the standard SUS score, add up all of the modified scores and then multiply the result by 2.5. A more efficient method of calculating a standard SUS score from a collection of raw item ratings is demonstrated by the following equation:

SUS=2.5(20+SUM(SUS01,SUS03,SUS05,SUS07,SUS09) − SUM(SUS02,SUS04,SUS06,SUS08,SUS10)) (Lewis, 2018)

In order to cover a variety of potential user reactions to various elements of system usability, Brooke (1996) (cited in Lewis, 2018) first developed 50 statements. He then assigned 20 members of his office's systems engineering team, each with a different background (secretary, systems programmer, etc.), to complete duties using two software programs, one of which was known to be reasonably simple to use and the other to be somewhat complex. In this very small-sample experiment, the 10 elements that were most effective for differentiating between simple and challenging applications were chosen to become the standard SUS. (Lewis, 2018)

According to Brooke (cited in Lewis, 2018), the correlations between the items had high absolute values (0.7–0.9). The results of this study led Brooke (1996) to conclude that SUS provides a single figure that is a composite indicator of the system's overall usefulness. Keeping in mind that individual item scores by themselves do not indicate anything. (Lewis, 2018)

Although the original study's content validity appears to have been useful, no metrics for construct validity, concurrent validity, or reliability were given—likely due to the sample size of only 20 being too small. However, Lucey (1991) (cited in Lewis, 2018) indicated in an unpublished thesis that the reliability of the SUS (coefficient alpha) was an acceptable 0.85 before Brooke (1996). (Lewis, 2018)

The flexibility of the System Usability Scale (SUS), which expands beyond small phrasing adjustments, is an additional benefit in real-world UX design. Researchers have recently extended their use beyond traditional usability testing to evaluate the perceived usability of items or product classes retrospectively (Grier, Bangor, Kortum, & Peres, 2013; Kortum & Bangor, 2013) (cited in Lewis, 2018). Grier (2013) (cited in Lewis, 2018) did not evaluate the SUS's psychometric qualities, but he did describe a version of the test that was modified for the purpose of purchasing easily maintained and troubleshooted items for the US military. (Lewis, 2018)

According to the study conducted by Lewis (2018) about SUS, SUS seems to have a highly promising future for usability and user experience research and practice. Although one might believe that interest in the SUS would have decreased with its age, studies on it are actually accelerating. Additionally, the body of research on SUS suggests that it is an effective tool for evaluating perceived usability. Nevertheless, there is still a great opportunity for more study to fill the current gaps. (Lewis, 2018)

However, despite its functionality and ease of use, Alkinani and Alzahrani (2021) reported in their study that no one has used it in the evaluation process related to Learning Management Systems, and mentioned that the criteria have barely been used. (Alkinani and Alzahrani, 2021)

4. Results and Data Presentation

4.1 System Usability Scale (SUS) Score Analysis

After performing several tasks in Google Classroom sections (Stream, Classwork, People, Grades), ratings were assigned and given based on active notes taken during the evaluation process. The results showed a high satisfaction when using the platform. The adjusted System Usability Scale score was 34 after the summation procedure. Producing 85/100. Which is considered Excellent (A) according to Bangor (2009) rating scale.

The main advantages of Google Classroom are as follows:

1. The overall navigation buttons are easy to find. And each button is labeled clearly according to its functions.

2. In the stream section, creating a new announcement (post) was made clear. However, there is no critical need to add a scrolling line to drag and move between pages, as there are no pages but the current one.

3. Class materials and supporting documents can be found easily in the classwork section. The student can navigate between materials without feeling lost.

4. The loading process is considered fast. For example, when clicking on any class material that contains a link to explore. It loads and opens fast.

5. People involved in the class can be recognized easily in the people's section. Students and teachers are categorized clearly so that the user can find the enrolled people effectively.

6. The navigation between graded items was easy to do, and grades can be displayed clearly. (Mhd Laith Alkhateeb, 2025)

The Full System Usability Scale questionnaire table and calculations will be provided in Appendix B.

4.2 Heuristic Evaluation Findings

After conducting a Heuristic evaluation on Google Classroom, the heuristic evaluation has identified several usability issues related to user control and freedom, consistency and standards, error prevention, recognition, flexibility and efficiency, and error recovery. The results were interpreted based on notes taken during the evaluation process.

The main disadvantages of Google Classroom are as follows:

1. There is no sign or an undo button when deleting a post.

2. There is no (Save all files offline) on the web version, unlike the mobile one.

3. In the classwork section, there is no instant downloading button when opening a class material file. The student has to choose a third-party system to download the material.

4. There is a Google Meet option in the stream section, while it actually exists in the navigation bar alongside labeled sections.

5. The student is not given a warning message when submitting late work.

6. When attempting to remove a student, an error message presents the issue in plain language. However, there is no solution provided to solve the issue. For

example, instead of presenting a solution, the message contained a phrase (try again later) (Mhd Laith Alkhateeb, 2025)

Full Heuristic Evaluation and Usability Issues Table will be provided in Appendix A

Supporting Documentation will be provided in Appendix C.

4.3 Detailed Usability Improvement Recommendations

The recommended practices to improve the usability of Google Classroom as an online learning platform are as follows:

1. An undo option should be added to easily restore the accidentally deleted.

2. (save files offline) option should be added in the web version to match the mobile version

3. A download button in the classwork section should be provided to download the class material effectively.

4. The redundant Google Meet button in the stream section should be removed to avoid confusion.

5. A warning message before the submission after the deadline should be considered for inclusion. For example, (you are about to submit a late work, the teacher may reduce the mark.s

6. Improve the clarity of error messages when they appear by explaining the cause and solutions. (Mhd Laith Alkhateeb, 2025)

5. Project Evaluation and Conclusion

5.1 Systematic Project Management and Reflection

Overall, by adhering to a clear schedule and project deliverables, this project followed an organized approach. A weekly action plan was used to schedule each stage, including the literature review, methodology development, the evaluation process, and reporting. The constant monitoring made sure that tasks were finished before the project timeline's deadline. Last-minute development and heavy work have been reduced by following this strategy. The effective organization of notes, results, and references was made easy by the use of digital tools like Google Docs and Google Keep.

There are advantages and disadvantages of doing a solo-based usability evaluation. The ability to conduct a comprehensive heuristic analysis without the requirement for participants was a major advantage that guaranteed complete control over the evaluation environment and saved time.

However, because the evaluation was based on one perspective without participants, this step could have created subjectivity. Consequently, a full heuristic evaluation table, which is based on Nielsen's 10 usability heuristics, was created along with violated principles and will be provided in Appendix A to reduce bias, and detailed notes were collected to objectively support each observation.

The project required careful scheduling and prioritization. The project was successfully finished within the assigned timeline. Although account limitations initially made it difficult to use several Google Classroom features, consequently, a personal account

was used in addition to the student account to clearly take notes, collect data, and evaluate from the perspectives of both students and teachers. Screenshots were taken without any problems during the evaluation procedure, and there were no technical problems, such as internet connection issues.

5.2 Conclusion

In summary, despite the limitations involved in this evaluation, such as the solo-based evaluation and limited accessible features in Google Classroom, which were addressed effectively, the project was considered a success as all objectives were addressed clearly, and the evaluation procedure was based on complete transparency.

The Heuristic Evaluation and the System Usability Scale have demonstrated great results regarding the usability of Google Classroom as an online learning platform. The SUS score was 85/100. This demonstrates the high satisfaction and the ease of use of the platform. While some usability heuristic principles were violated, such as user control and freedom, consistency and standards, error prevention, recognition, flexibility and efficiency, and error recovery. The platform has shown a consistent layout. The easy navigation, clear labeling, and fast loading have made the platform a great choice for learners and teachers. Making it a top choice for educational institutions to adopt, implement, and use to ensure a happy learning environment.

6. References and Appendices

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6.2 Appendix A: Full Heuristic Evaluation and Usability Issues Table

Nielsen’s Severity Rating (Paramitha, Dantes and Indrawan, 2018)

Rating	Definition
0	No usability problem at all
1	Aesthetic problem that requires an easy fix if time is available
2	Minor usability issue: fixing is of low priority
3	Major usability problem and fixing requires high priority

4	Usability catastrophe: needs to be resolved prior to product release

Nielsen's Severity Rating (Paramitha, Dantes and Indrawan, 2018)

#	Nielsen's Heuristic Principles	Number of Issues
1	Visibility of system status	
2	Match between the system and the real world	
3	User control and freedom	1

4	Consistency and standards	1
5	Error prevention	1
6	Recognition rather than recall	1
7	Flexibility and efficiency of use	1
8	Aesthetic and minimalist design:	
9	Help users recognize, diagnose, and recover from errors	1
10	Help and documentation	
	Total Problems Found	**6**

Heuristics Evaluation Results (Mhd Laith Alkhateeb). The table layout, including columns and naming used by Paramitha, Dantes and Indrawan (2018)

#	Violated principle	Severity Rating	Recommendations
1	**User control and freedom:** No undo when deleting a post in the stream section	3	Add an undo option to restore the accidentally deleted easily

2	**Consistency and Standards/Flexibility and Efficiency of Use** The web version of GC does not contain (save files offline), unlike the mobile version	2	Add (save files offline) in the web version to match the mobile version
3	**Flexibility and Efficiency of Use/Recognition rather than recall** There is no instant download button in the classwork section. Students must use a third-party system to download	3	Provide a download button in the classwork section to download the class material effectively
4	**Consistency and Standards** The Google Meet join button appears redundant in the stream section and the navigation bar	2	Remove the redundant Google Meet button in the stream section to avoid confusion
5	**Error prevention/Help users** **recognize, diagnose, and recover** **from errors** There is no warning from the student's perspective when submitting late work	3	Include a warning message before the submission after the deadline. For example, (you are about to submit a late work, the teacher may reduce marks.
6	**Help users recognize, diagnose,** **and recover from errors**	3	Improve the clarity of error messages when they appear by explaining the cause and solutions

	When removing a student, an error message appears saying (try again later) instead of proposing a solution		

6.3 Appendix B: System Usability Scale Questionnaire table and calculations

SUS Questionnaire Statements (Brooke, 1996)

#	System Usability Scale Questionnaire Statements
1	I think that I would like to use this system frequently
2	I found the system unnecessarily complex

3	I thought the system was easy to use
4	I think that I would need the support of a technical person to be able to use this system
5	I found the various functions in this system were well integrated
6	I thought there was too much inconsistency in this system
7	I would imagine that most people would learn to use this system very quickly
8	I found the system very cumbersome to use
9	I felt very confident using the system
10	I needed to learn a lot of things before I could get going with this system

SUS Grades Classification (Abushamleh and Jusoh, 2021)

SUS Score	Grade	Adjective Rating
80-100	A	Excellent
68-79	B	Good
50-67	C	Okay
Less than 50	F	Weak

SUS Raw Ratings Table (Mhd Laith Alkhateeb, 2025)

#	SUS Questionnaire Statements	Rating (raw (1-5))	Justification
1	I think that I would like to use this system frequently	4	Easy navigation, clear labels, and fast loading
2	I found the system unnecessarily complex	2	Disagree, because navigation and labeling are clear. However, there

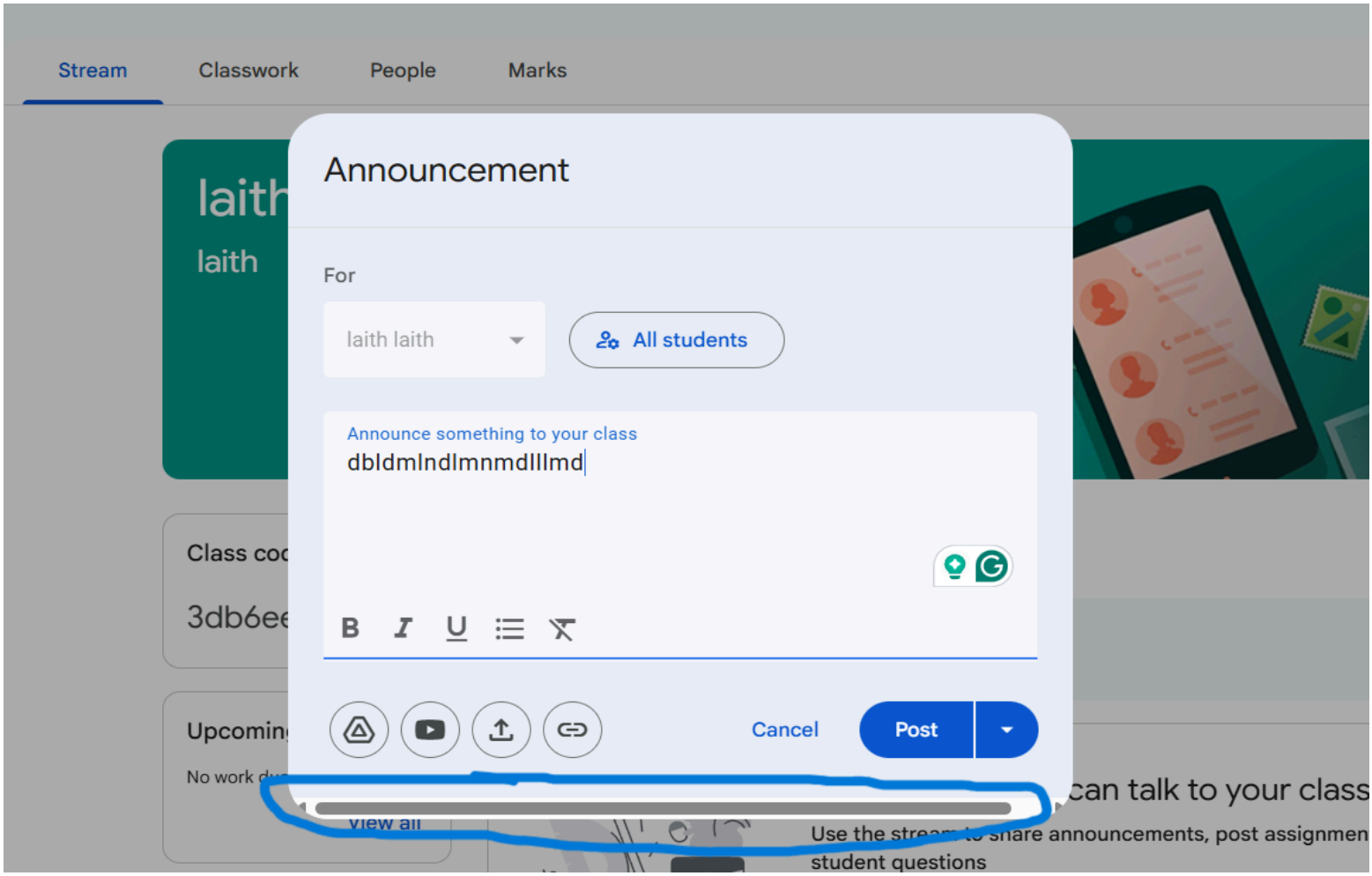
			is some small complexity in the downloading behavior
3	I thought the system was easy to use	5	Easy navigation and clear classwork organization
4	I think that I would need the support of a technical person to be able to use this system	1	Strongly disagree, because tasks are straightforward
5	I found the various functions in this system were well integrated	4	Navigation, such as classwork, people, and grades, can all be reached and organized logically
6	I thought there was too much inconsistency in this system	2	Consistent overall. However, there are a few inconsistencies in the web and mobile versions (differences in saving)
7	I would imagine that most people would learn to use this system very quickly	5	Easy and quick to learn. Because of the easy navigation and clear labels
8	I found the system very cumbersome to use	1	Strongly disagree. Because loading is fast and navigation between graded items is easy
9	I felt very confident using the system	4	The level of confidence is high because of the clarity and clear categorization of sections
10	I needed to learn a lot of things before I could get going with this system	2	Disagree. Because the interface is clear and straightforward

Adjusted score = for odd numbers, subtract 1 from the rating. For even numbers, subtract the rating from 5

Final adjusted score = 34

SUS score = 34 * 2.5 = 85/100

6.4 Appendix C: Supporting Documentation



This picture shows the scrolling icon appears when making announcements, which is unnecessary.

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L5DC-BM

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Business IT Project complete class ma... Posted Oct 19

Business IT Project complete class material for L5DC-BM

BITP Topic 1 Introduction.... Microsoft PowerPoint
BITP Topic 1 Introduction.pptx
BITP Topic 2 Conduting R... Microsoft PowerPoint
BITP Topic 3 Methods of S... Microsoft PowerPoint
BITP Topic 4 Testing and ... Microsoft PowerPoint
BITP Topic 5 Planning the ... Microsoft PowerPoint

View material

Business Project Managment Posted Sep 30

siness IT Porject Posted Sep 25

ogle.com/file/d/1kBQO9KPKsdaONVboBZ9y9wXA-TDzTKQS/view?usp=classroom_web&authuser=0

Stream Classwork People

Business IT Project

L5DC-BM

Meet

Join

Upcoming

Woohoo, no work due soon!

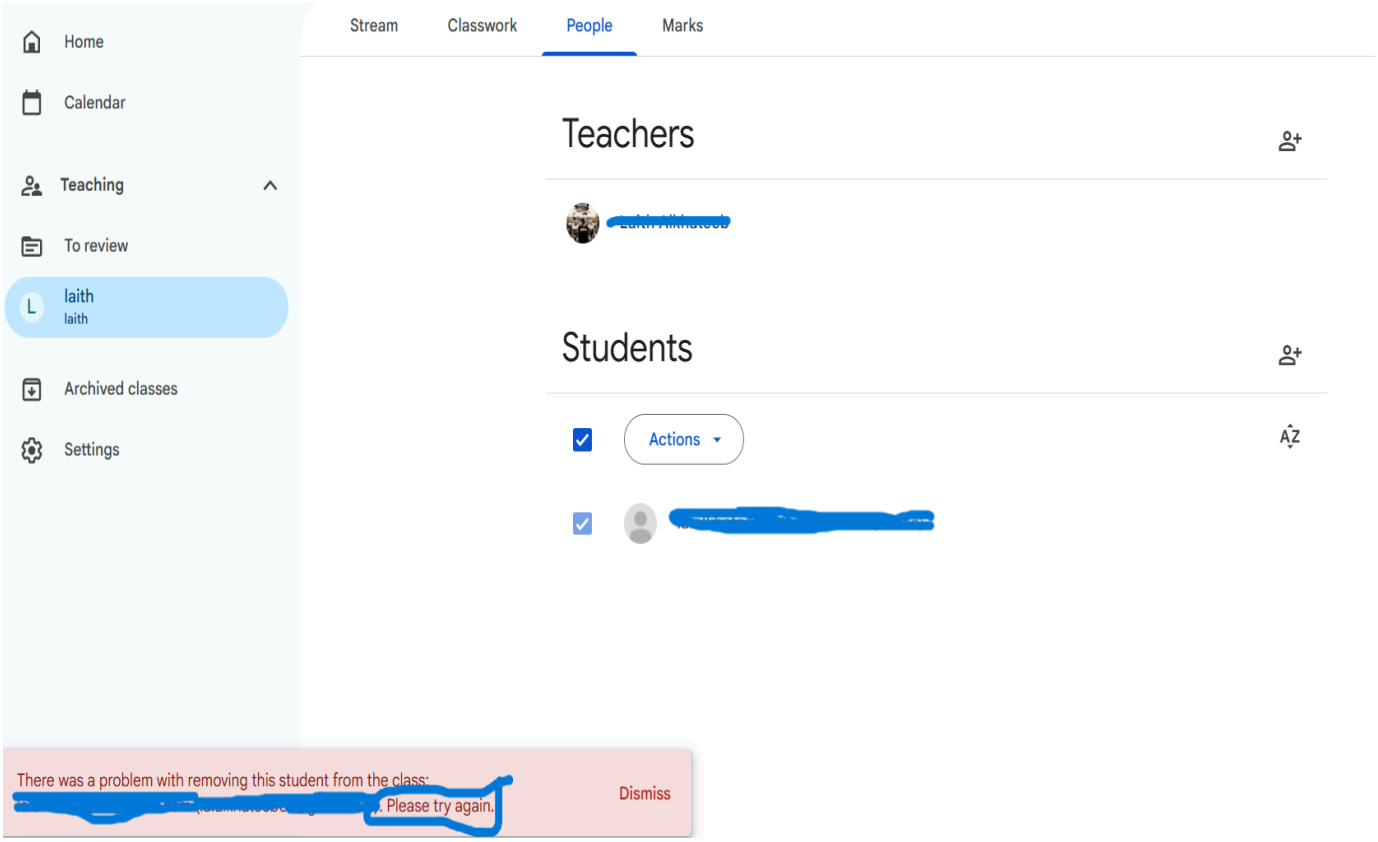
View all

New announcement

Muhammad Rafiq posted a new material: Business IT Project complete class material for L5...
Oct 19

TNEI Academics posted a new material: Business Project Managment
Sep 30

Muhammad Rafiq posted a new material: Business IT Porject
Sep 25



this picture shows how vague the error message is without clearly stating the cause and the solution

This picture illustrates the lack of instant download button when attempting to download the file

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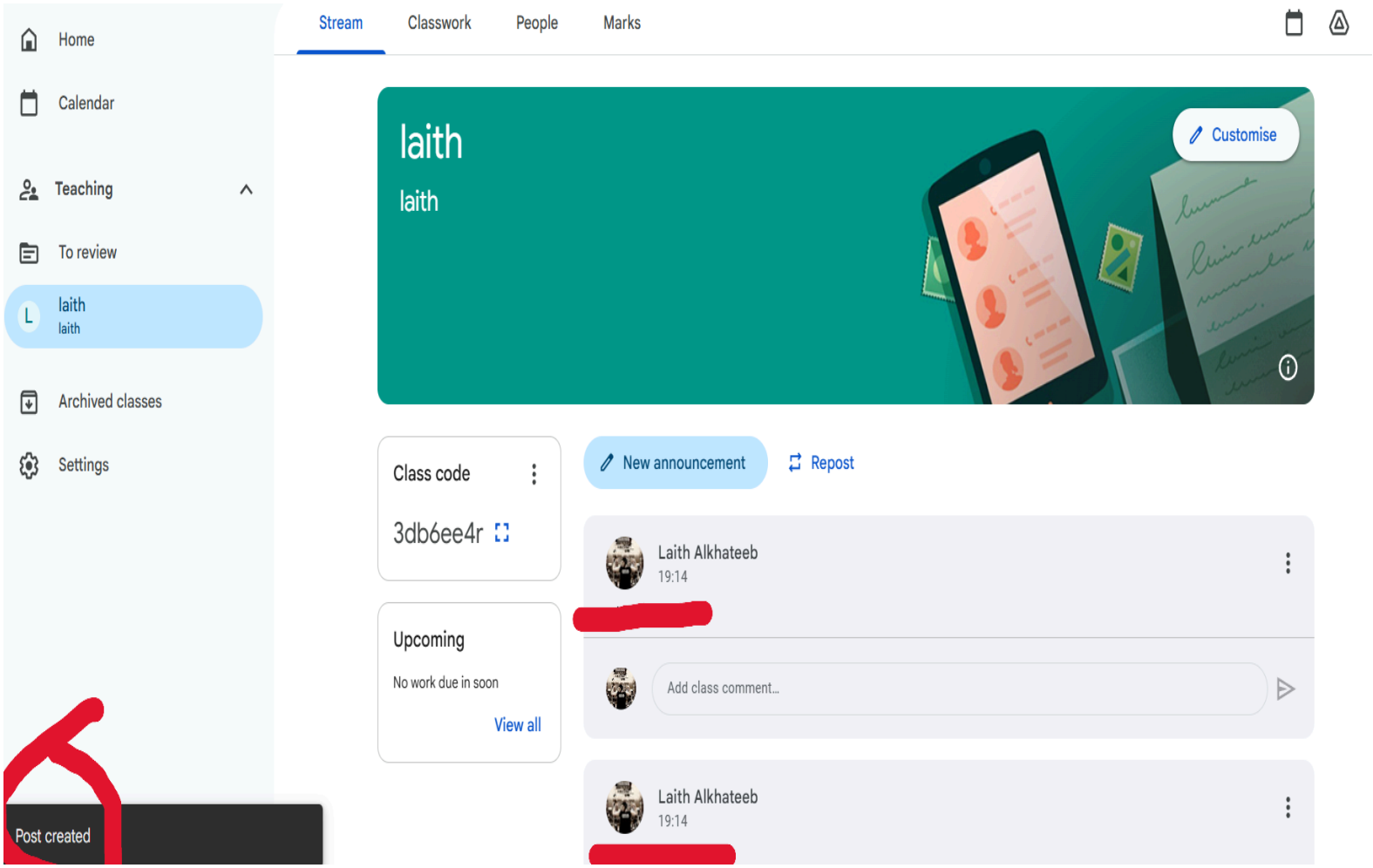

This picture illustrates the unexistance of an undo button when creating a post

Evaluating the usability of Google Classroom as an online learning portal for universities (Project Presentation)

The Rise and Challenge of Online Learning Platforms

- Online Learning Platforms are now vital tools in higher education they participate in enhancing communication and streamlining resource management.
- Google Classroom is one of the most widely used Online Learning Platforms due to its seamless integration with Google Workspace and cross-device accessibility.
- Despite its widespread use, Google Classroom still has usability problems. For example, both students and teachers have reported navigational difficulties.

The Core Problem: Usability Gap in Google Classroom

- Although students had given high rates to the platform for its usability and mobile-friendliness, several issues have been identified.
- Teachers reported that the interface is somewhat complicated and lacks user-friendliness, potentially affecting the platform's overall efficiency.
- Previous studies found that although Google Classroom is valued for its organisation, other important components were criticised, such as the stream interface's dull, non-personalised user interface, and its annoying scrolling.
- The primary goal of the study is to assess Google Classroom usability in order to identify certain user problems and offer beneficial recommendations.

Project Aims and Objectives

- **Aim**: To evaluate Google Classroom usability as an online learning platform for universities from a Human-Computer Interaction (HCI) perspective.
- **Objective 1**: To perform a quantitative assessment of user satisfaction and perceived usability using the System Usability Scale (SUS).
- **Objective 2**: To perform a qualitative assessment of the interface design to identify specific usability issues using Jakob Nielsen's Heuristic Evaluation.
- **Objective 3**: Provide comprehensive and practical recommendations for enhancing Google Classroom user interface and general user experience.

Methodology: Mixed Methods Approach

- **Approach**: Mixed methods (Qualitative and Quantitative).
- Methods Used: System Usability Scale (SUS) and Heuristic Evaluation Method.
- These methods are reliable, easy to implement, and are commonly used in HCI research to assess usability.
- **Process**: A solo evaluation process was conducted, primarily from a student's perspective, due to the unavailability of participants.
- **Heuristic Criteria**: A refined version of Jakob Nielsen's Usability Heuristics, adapted for Learning Management Systems by Mtebe and Kissaka (2015), was applied

Quantitative Results: System Usability Scale Score

- **Final Adjusted Score**: 34
- **System Usability Scale (SUS) Score**: 85/100
- **Adjective Rating (Bangor, 2009)**: Excellent (A)
- **Conclusion**: The quantitative result demonstrates high satisfaction and perceived ease of use for the platform.

Qualitative Results: Key Usability Strengths

- **Visibility of System Status**: Fast loading times for class materials and links.
- **Match between System and Real World**: Clear and easy-to-find navigation buttons with labels matching their functions.
- **Recognition rather than Recall**: Effective categorization of enrolled people (Teachers and Students) in the 'People' section.
- **Consistency and Standards**: The overall layout is consistent, contributing to easy navigation.
- **Efficiency**: The navigation between graded items is straightforward and grades are displayed clearly.

Qualitative Results: Usability Issues (Violations)

- **User Control & Freedom**: No 'undo' button when deleting a post (Severity 3 - Major Problem).
- **Consistency & Standards**: The 'save files offline' option exists on the mobile version but is absent on the web version (Severity 2 - Minor Issue).
- **Flexibility & Efficiency of Use**: No instant download button for class material files, forcing users to use a third-party system (Severity 3 - Major Problem).
- **Error Prevention**: No warning message is given to the student before submitting late work.
- **Error Identification & Recovery**: When a student is removed, for example, error messages lack a helpful solution and use vague phrases like "try again later" (Severity 3-Major Problem).

Recommendations for Improvement

- **Improve User Control**: Implement an 'Undo' option to easily restore accidentally deleted content.
- **Enhance Consistency**: Add the 'save files offline' function to the web version to match the mobile application.
- **Increase Efficiency**: Provide a direct download button within the classwork section for all class materials.
- **Implement Error Prevention**: Include a clear warning message before late submission, such as, "You are about to submit a late work, the teacher may reduce the marks.".
- **Refine Error Handling:** Improve error messages to clearly explain the cause of the problem and suggest a solution.
- **Remove Redundancy**: Remove the redundant Google Meet button from the stream section as it already exists in the navigation bar.

Critique: What Went Well (Systematic Project Management)

- **Organized Approach**: The project followed a clear schedule and a weekly action plan, allowing all stages from literature review to reporting to be finished on time
- **Efficient Data Management**: Using digital tools (Google Docs/Keep) made it easier to organize notes, findings, and references, which decreased the amount of work that needed to be done at the last minute.
- **Comprehensive Qualitative Analysis**: The solo heuristic evaluation was finished completely, ensuring complete control over the process and reducing the time needed to gather participants.
- **Objective Support**: Detailed notes were collected and a complete heuristic evaluation table was created to objectively support each observation and minimize bias.

Critique: What Didn't Go Well (Limitations & Challenges)

- **Solo Evaluation**: The main limitation was the reliance on a single perspective, which could potentially introduce subjectivity to the findings
- **Account Limitations**: Initial limitations with the student account made it difficult to access all Google Classroom features, requiring the use of a secondary personal account to cover both student and teacher perspectives.

Learning and What to Do Differently Next Time

- **Learned**: The importance of performing both quantitative and qualitative assessments to gain a comprehensive view of perceived and actual usability
- **Learned**: How to apply HCI principles and severity ratings to objectively classify and prioritize identified usability issues.
- **Do Differently**: Involve a group of participants such as students and teachers to perform the SUS survey and conduct the Heuristic Evaluation more effectively.
- **Do Differently**: Implement various usability analysis methodologies and protocols in the qualitative phase to observe how users navigate and interact with the interface in real-time.

Project Conclusion

- The SUS score of 85/100 confirms the high satisfaction and ease of use of Google Classroom.
- The Heuristic Evaluation identified 6 specific issues, primarily related to User Control, Efficiency, and Error Recovery.
- The platform is a strong choice for educational institutions due to its consistent layout, clear navigation, and fast performance.
- **Final Recommendation**: Implement the suggested improvements to address the critical heuristic violations and significantly enhance the overall user experience and teaching efficiency.

AI Acknowledgment

This report was developed by the student. AI tools such as ChatGPT and Gemini LLM were used to assist with research organization, idea generation. Grammarly was used for proofreading and grammar correction. All data, interpretations, writing, and conclusions are original.